

2. Sustainable Facility and Estate Intelligence Dashboard for India

Problem Statement:

Create an AI-powered decision dashboard for government-run and institution-run facilities across India that helps analyze sustainability, safety, and operations challenges such as air quality, waste management, water usage, energy consumption, traffic flow, asset utilization, emissions, and climate resilience. The solution should be adaptable to hospitals, colleges, company campuses, public-sector units, industrial estates, municipal facilities, and large mixed-use institutional zones.

Description - Expected Features:

The dashboard should ingest open, synthetic, or sensor-style data; visualize trends; detect anomalies; forecast short-term changes; and generate plain-language recommendations for administrators. Example modules may include air-quality monitoring, waste-bin overflow prediction, energy-usage optimization, water-use alerts, traffic or parking hotspot analysis, equipment utilization, safety incident trend monitoring, and sustainability scorecards. A strong prototype should combine data visualization with AI-generated insights, scenario simulation, and action recommendations such as changing collection routes, sending advisories, adjusting energy schedules, identifying pollution hotspots, optimizing facility maintenance, or prioritizing inspection zones. Teams may frame the problem for a government hospital, engineering college, municipal campus, industrial estate, company campus, or public-sector manufacturing facility.

Evaluation Criteria / Weightage:

Sustainability and operations relevance across Indian facilities	20%
Data engineering and visualization	20%
AI/ML modeling, forecasting, or anomaly detection	20%
Actionability of recommendations	15%
Usability for facility administrators	10%
Explainability and assumptions	10%
Demo Quality	5%

Guidelines:

Prefer Python, pandas, scikit-learn, Prophet or other forecasting libraries, Streamlit, Power BI-style dashboards, geospatial libraries, map APIs, IoT simulators, and GenAI-based insight generation. Teams should use public datasets, synthetic IoT feeds, or manually prepared sample data. Include assumptions behind forecasts and recommendations. If using environmental, safety, or health-impact data, avoid overstating conclusions; present outputs as decision-support insights, not official measurements. Solutions should be configurable for different facility types and should support role-based views for administrators, operations teams, and sustainability officers.